

Design and Simulation of a Solar Generation Plant with Battery Storage for Residential Supply

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Abstract—This paper presents the design and simulation of an isolated photovoltaic generation plant with battery storage for residential loads. The proposed system combines a photovoltaic array, a DC/DC Boost converter for maximum-power operation and DC-bus regulation, a bidirectional Cuk converter for battery charge and discharge, and a three-phase DC/AC inverter for supplying variable household demand. A 20-panel photovoltaic array was selected to provide approximately 10 kW at peak irradiance, while a 450 V DC bus was adopted as the interface with the inverter. The system was evaluated in PSIM under different irradiance levels and load intervals representing a full day of operation. The simulation results show that the photovoltaic array reaches the required generation level, the DC bus remains close to the specified value despite small fluctuations, and the inverter produces phase voltages near 127 V_{RMS}. The load-current results agree closely with theoretical values, with deviations of approximately 3% or less. These results validate the feasibility of the proposed residential stand-alone photovoltaic system and identify future improvements related to automatic MPPT control and enhanced DC-bus stabilization.

Index Terms—photovoltaic generation, battery storage, Boost converter, bidirectional Cuk converter, three-phase inverter, residential loads, PSIM simulation.

I. INTRODUCTION

The growing need for clean and reliable residential energy systems has increased the relevance of isolated photovoltaic (PV) generation plants with local storage. In these systems, solar panels produce DC power, power-electronic converters regulate voltage and current, batteries provide energy availability when solar irradiance is insufficient, and inverters supply AC loads.

This work redesigns and simulates a residential stand-alone solar generation plant with battery support. The system is intended to supply variable household demand during a full day while maintaining a regulated DC bus and acceptable AC output voltage. The main contributions are: the sizing of the PV array, the calculation of converter parameters, the design of a bidirectional battery interface, and the evaluation of the inverter and load behavior under changing irradiance and demand.

II. PROBLEM DESCRIPTION

The objective is to perform a basic design of a stand-alone solar generation plant for a residence. The plant must deliver energy to the household load during both day and night. For this purpose, a set of commercial solar panels must be selected so that the maximum available energy delivered to

the home electrical network is 10 kW at noon under clear-sky conditions. The required power converters must be designed for the application, an appropriate battery or battery bank must be selected for energy storage, and the converter needed for connection to the household electrical network must be designed. The residential network is considered three-phase.

The residence is modeled as a variable load throughout the day. Based on three-phase current measurements, the load demand is defined as follows: from 6:00 a.m. to 11:00 a.m., the residence consumes a total of 15 A, equivalent to 5 A per phase; from 11:00 a.m. to 4:00 p.m., it consumes 60 A; from 4:00 p.m. to 10:00 p.m., it consumes 30 A; and from 10:00 p.m. to 6:00 a.m., it consumes 1 A.

III. LITERATURE REVIEW

A. Photovoltaic Systems

Photovoltaic systems generate electrical energy from solar radiation by means of solar panels. Panels can be connected in series, parallel, or mixed series-parallel arrangements. Series connections increase voltage while maintaining current, whereas parallel connections increase current while maintaining voltage [2]. To maximize extracted energy, PV arrays should operate close to their maximum power point; this is commonly achieved through maximum power point tracking (MPPT) controllers [3]. Bypass diodes are also used to reduce the effect of partial shading or damaged cells by allowing current to flow around affected sections of the panel [4].

B. DC/DC Power Converters

DC/DC converters regulate an output voltage from a given DC input by controlling semiconductor switching through pulse-width modulation (PWM) [5]. A Buck converter reduces voltage, a Boost converter increases voltage, and a Cuk converter can operate as a step-up/step-down topology with inverted output polarity. The Cuk converter uses inductors and capacitors for input and output filtering and transfers energy mainly through an intermediate capacitor [8]. A bidirectional Cuk converter replaces the diode with an actively controlled MOSFET, enabling power flow in both directions and making it suitable for battery charge and discharge applications [9].

C. Battery Storage

Battery storage is required in isolated PV systems to maintain supply when solar radiation is unavailable. A PV battery charger typically includes a PV generator, a power converter,

TABLE I
MAIN CHARACTERISTICS OF THE SELECTED PV PANEL

Specification	YS700M-66
Maximum power, P_{max}	700 W
Maximum power voltage, V_{mp}	39.58 V
Maximum power current, I_{mp}	17.69 A
Open-circuit voltage, V_{oc}	46.75 V
Short-circuit current, I_{sc}	18.92 A
Operating temperature	-40 °C to 85 °C
Maximum series fuse rating	25 A
Power tolerance	0-3%
Temperature coefficient of P_{max}	-0.35%/°C
Temperature coefficient of V_{oc}	-0.29%/°C
Temperature coefficient of I_{sc}	0.048%/°C

a battery bank, and a control system [10]. In the proposed design, a bidirectional Cuk converter interfaces the battery with the DC bus so that the battery can absorb excess PV energy or inject energy when the PV source is insufficient.

D. Three-Phase Voltage Inverter

An inverter converts DC voltage into AC voltage. A three-phase inverter can be formed by three half-bridge legs operating in parallel, each controlled by sinusoidal reference signals displaced by 120°. The references are compared with a triangular carrier signal to generate PWM gate pulses. The amplitude modulation index controls the output voltage, while the frequency modulation index determines the carrier frequency and affects harmonic content.

IV. SYSTEM DESCRIPTION

A. Photovoltaic Array

The required number of panels can be estimated from

$$N_{panels} = \frac{P_{required}}{P_{panel}}. \quad (1)$$

For the residential isolated system, a nominal 700 W panel was selected. Its main specifications are summarized in Table I.

B. Boost Converter for the DC Bus

A Boost converter was selected to raise and regulate the PV-array voltage at the input of the DC bus. The target bus voltage was set to 450 V. The converter duty ratio and passive components were calculated using the following equations:

$$d_{Boost} = 1 - \frac{V_{in}}{V_o}, \quad (2)$$

$$i_{in} = i_L, \quad (3)$$

$$i_o = i_{in}(1 - d_{Boost}) = \frac{P_o}{V_o}, \quad (4)$$

$$L = \frac{V_{in}d_{Boost}}{f_{Boost}\Delta i_L}, \quad (5)$$

$$C = \frac{i_o d_{Boost}}{f_{Boost}\Delta V_o}. \quad (6)$$

C. Bidirectional Cuk Converter for Batteries

A bidirectional Cuk converter was used for battery energy management because it supports power flow in both directions and provides input/output filtering. Its main design equations are:

$$d_{Cuk} = \frac{V_o}{V_{in} + V_o}, \quad (7)$$

$$V_{C1} = \frac{V_{in}}{1 - d_{Cuk}} = \frac{V_o}{d_{Cuk}}, \quad (8)$$

$$i_{in} = i_{L1} = i_o \left(\frac{d_{Cuk}}{1 - d_{Cuk}} \right), \quad (9)$$

$$i_o = i_{L2}, \quad (10)$$

$$L_1 = \frac{V_{in}d_{Cuk}}{f_{Cuk}\Delta i_{L1}}, \quad (11)$$

$$L_2 = \frac{V_o(1 - d_{Cuk})}{f_{Cuk}\Delta i_{L2}}, \quad (12)$$

$$C_1 = \frac{i_{L2}d_{Cuk}}{f_{Cuk}\Delta V_{C1}}, \quad (13)$$

$$C_2 = \frac{V_o(1 - d_{Cuk})}{8f_{Cuk}^2 L_2 \Delta V_{C1}}. \quad (14)$$

D. Three-Phase DC/AC Inverter

The inverter supplies AC loads from the regulated DC bus. The phase-to-neutral voltage is determined by

$$V_{AN} = V_{available}m_a, \quad (15)$$

where the available voltage for a three-phase half-bridge inverter is

$$V_{available} = \frac{V_{input}}{2}. \quad (16)$$

The frequency modulation index is

$$m_f = \frac{f_t}{f_c}. \quad (17)$$

For this design, $m_f = 63$ was selected. With a 50 Hz system frequency, the triangular carrier frequency is 3150 Hz. Since the DC input is 450 V, the available voltage is 225 V. A phase voltage of 127 V_{RMS} corresponds to approximately 179.6 V peak; therefore, the amplitude modulation index is approximately 0.8.

V. SIMULATION DEVELOPMENT

A. Photovoltaic Farm

The selected panel was evaluated in PSIM. At 1000 W m⁻², the simulated maximum power point was 509.33 W, at 31.38 V and 16.23 A. The resulting panel power curve is shown in Fig. 1. To reach 10 kW at peak irradiance, 20 panels were required. The array was configured as five panels in series and four strings in parallel.

The power generated by the complete photovoltaic farm is shown in Fig. 2.

The PV farm was simulated from 250 W m⁻² to 1000 W m⁻² and back down to represent daily irradiance variation. This irradiance profile was implemented through step functions activated at different simulation times, so each step changes the

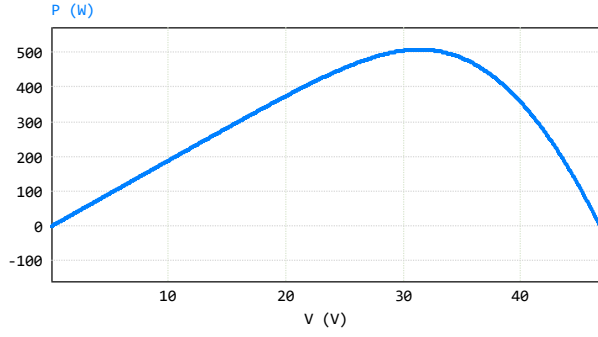


Fig. 1. Solar Panel PV curve

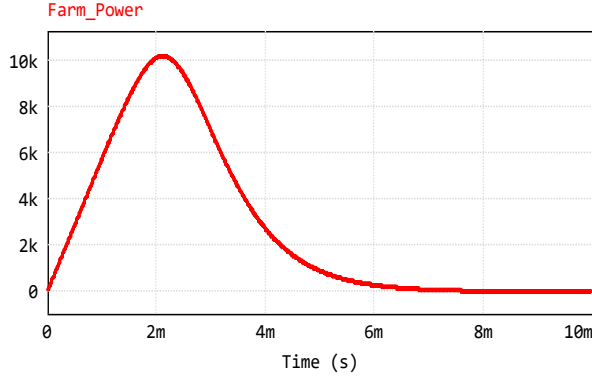


Fig. 2. Power generated by the photovoltaic farm

TABLE II
GENERATED PV-FARM VALUES DURING THE DAY

Irradiance (W/m ²)	Voltage (V)	Current (A)	Power (W)
250	153.45	16.64	2553.408
500	158.55	33.16	5257.518
750	158.65	49.28	7818.272
1000	156.95	64.92	10189.194

irradiance level according to the scaled daily schedule rather than using a separate figure. Table II summarizes the generated values.

B. Boost Converter

The Boost converter was designed to keep the output voltage at 450 V for each PV input condition. The duty ratios are shown in Table III, and the resulting passive-component values are shown in Table IV. The converter schematic is shown in Fig. 3. As with the irradiance profile, duty-cycle variation was simulated through step functions activated at different times; each step applies the duty ratio required for the corresponding irradiance interval, so no additional duty-cycle figure is included.

C. Bidirectional Cuk Converter

The Cuk converter was configured with a 60 V battery input and a 450 V output target. Applying the design equation gives

TABLE III
BOOST CONVERTER DUTY RATIOS

V_{in} (V)	V_o (V)	Duty ratio
153.45	450	0.6590000
158.55	450	0.6476667
158.65	450	0.6474444
156.95	450	0.6512222

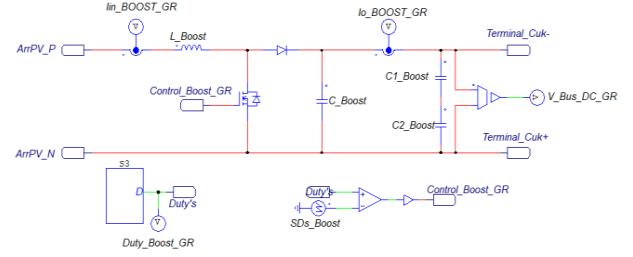


Fig. 3. Boost converter implemented in PSIM

TABLE IV
BOOST CONVERTER INDUCTANCE AND CAPACITANCE

Duty ratio	Inductance (H)	Capacitance (F)
0.6590000	30.39 m	187 μ
0.6476667	15.48 m	378.3 μ
0.6474444	10.40 m	562.4 μ
0.6512222	7.872 m	737.3 μ
Maximum	30.39 m	737.3 μ

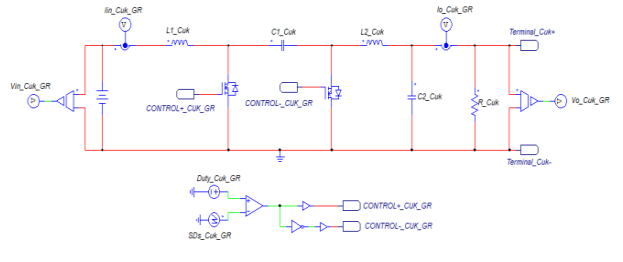


Fig. 4. Bidirectional Cuk converter implemented in PSIM

TABLE V
CUK CONVERTER INDUCTANCE AND CAPACITANCE

Power (W)	L_1	L_2	C_1	C_2
2553.408	622 μ H	466.5 μ H	2.454 μ F	3.94 nF
5257.518	302.1 μ H	226.6 μ H	5.053 μ F	8.113 nF
7818.272	203.1 μ H	152.4 μ H	7.515 μ F	12.07 nF
10189.194	155.9 μ H	226.6 μ H	9.794 μ F	15.72 nF
Maximum	622 μ H	466.5 μ H	9.79 μ F	15.72 nF

a duty ratio of 0.882352941. The PSIM implementation is shown in Fig. 4, and the calculated passive components for the possible power levels are listed in Table V.

D. Load Schedule

The daily residential load was represented by scaled simulation intervals. A full day was mapped to 0.96 s, so 1 h corresponds to 0.04 s. The inverter circuit, PWM generation, and

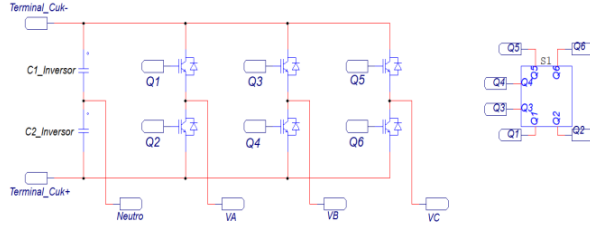


Fig. 5. Three-phase inverter implemented in PSIM

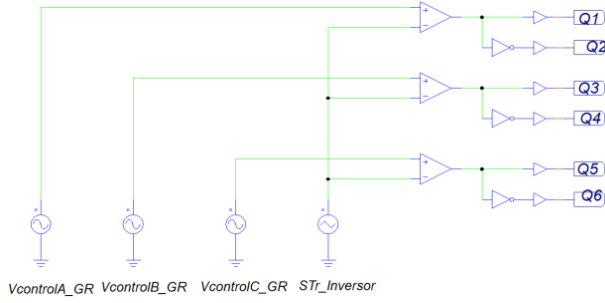


Fig. 6. PWM generation for the three-phase inverter

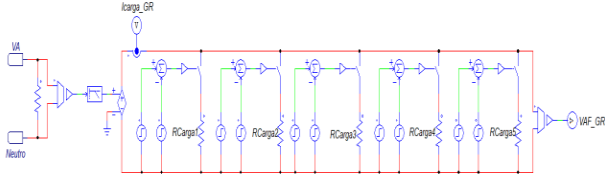


Fig. 7. Daily load-connection scheme

TABLE VI
LOAD CONNECTION TIMES

Current per phase (A_{RMS})	Day period	Scaled period
5	06:00–11:00	0.24–0.44 s
30	11:00–16:00	0.44–0.64 s
10	16:00–22:00	0.64–0.88 s
0.333	22:00–06:00	0.88–0.96 s

TABLE VII
LOAD RESISTANCE FOR EACH TIME INTERVAL

Period	V_{max} (V)	I_{max} (A)	R (Ω)
00:00–06:00	179.60	0.471	381.05
06:00–11:00	179.60	7.071	25.40
11:00–16:00	179.60	28.284	6.35
16:00–22:00	179.60	14.142	12.70
22:00–24:00	179.60	0.471	381.05

daily load-connection scheme are shown in Figs. 5, 6, and 7, respectively. The load intervals and equivalent resistances are presented in Tables VI and VII.

TABLE VIII
TIME SCALING FOR IRRADIANCE AND DUTY-CYCLE CHANGES

Day time	Simulation time (s)	Irradiance (W/m^2)
06:00	0.24	250
08:00	0.32	500
10:00	0.40	750
12:00	0.48	1000
14:00	0.56	750
16:00	0.64	500
18:00	0.72	250

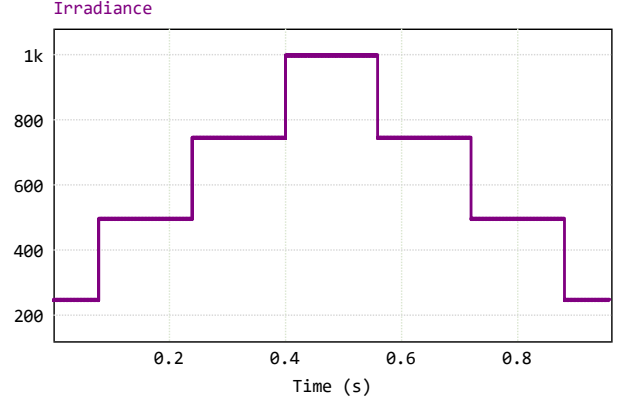


Fig. 8. Irradiance signal over the simulated day

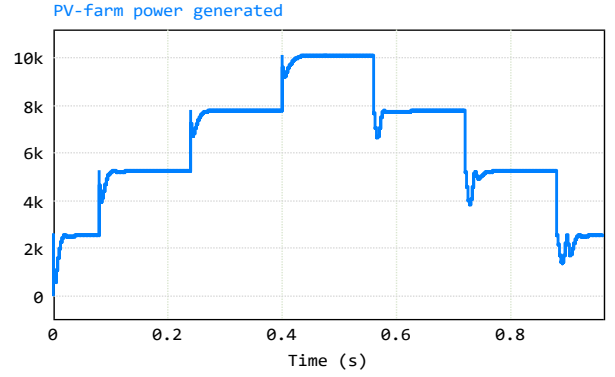


Fig. 9. PV-farm generated power as a function of irradiance

VI. RESULTS AND DISCUSSION

A. Photovoltaic Array Results

The irradiance signal followed the expected daily profile, increasing from $250 W m^{-2}$ to $1000 W m^{-2}$ and later decreasing, as shown in Fig. 8. The PV farm reached approximately 10kW at maximum irradiance; Fig. 9 shows the generated power, confirming that the 20-panel arrangement is suitable for the design objective.

Figures 10 and 11 present the photovoltaic-farm voltage and current separately during the simulation. These two plots are evaluated independently because the voltage remains comparatively close to the operating region of the array, while the current changes more directly with irradiance.

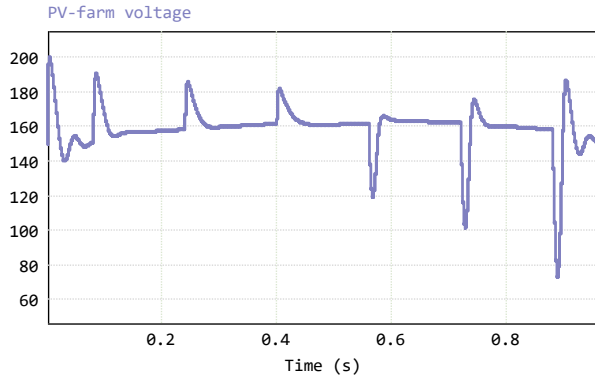


Fig. 10. PV-farm voltage during the simulation

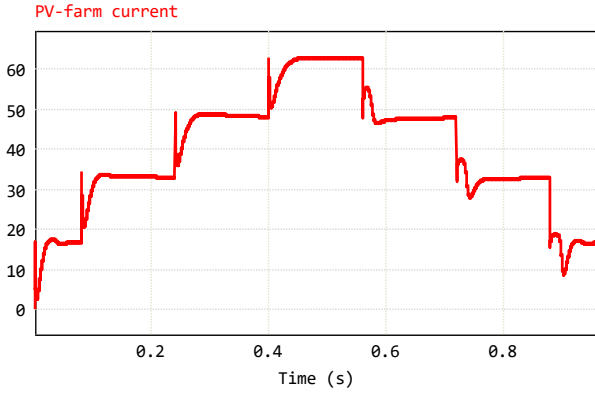


Fig. 11. PV-farm current during the simulation

TABLE IX
SIMULATED PV-FARM VOLTAGE AND CURRENT

Irradiance (W/m ²)	Voltage (V)	Current (A)
250	153.45	16.64
500	158.55	33.16
750	158.65	49.28
1000	156.95	64.92

Table IX compares the simulated PV voltage and current values obtained for each irradiance level.

B. Boost Converter Results

The Boost converter controlled the PV operating voltage by manually changing the duty cycle according to the irradiance condition. The current at the converter input followed the expected PV-array behavior, as shown in Fig. 12, while the output voltage remained close to the 450 V DC-bus target, as shown in Fig. 13. Small disturbances appeared during duty-cycle transitions, indicating that an automatic MPPT or closed-loop control strategy could further improve performance.

C. Bidirectional Cuk Converter Results

The battery was represented by a 60 V input source, as shown in Fig. 14. The Cuk converter inverted the voltage polarity and generated an output near -450 V, as shown in Fig. 15. When connected with reversed polarity to the DC

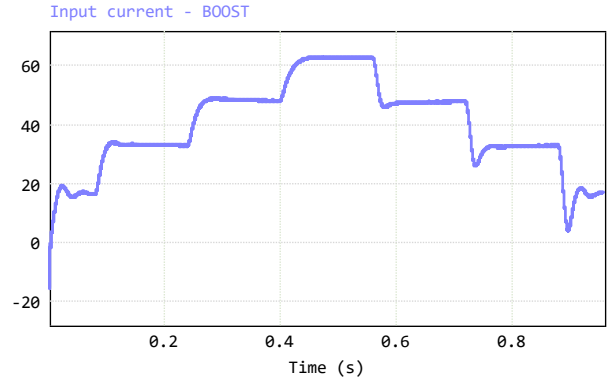


Fig. 12. Input current of the Boost converter

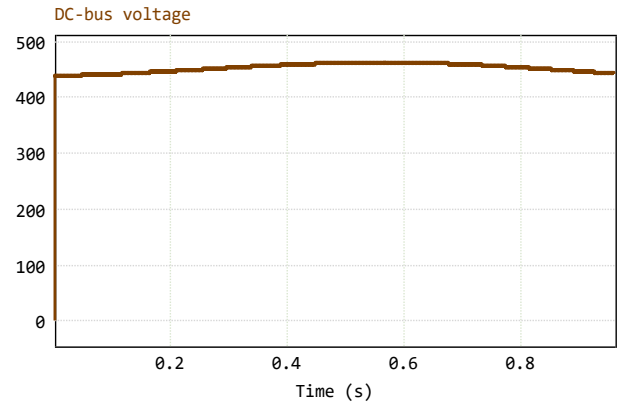


Fig. 13. DC-bus voltage

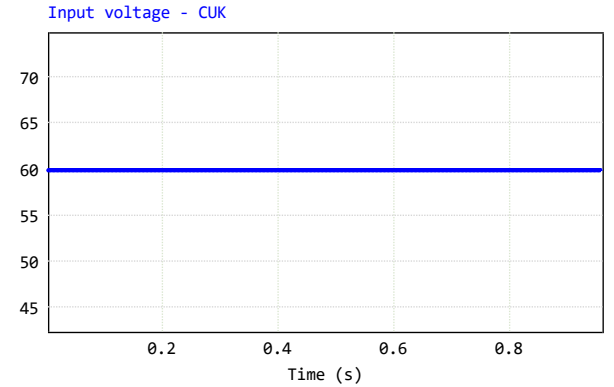


Fig. 14. Battery/input voltage of the Cuk converter

bus, this behavior provided the desired positive bus voltage. The output presented small fluctuations, which propagated to the inverter stage but remained within an acceptable range for the load analysis.

D. Three-Phase Inverter Results

The inverter generated three phase-to-neutral voltages displaced by 120° , as shown in Fig. 16. The RMS values were slightly higher than the expected 127 V, mainly due to DC-bus fluctuations. Table X shows the comparison.

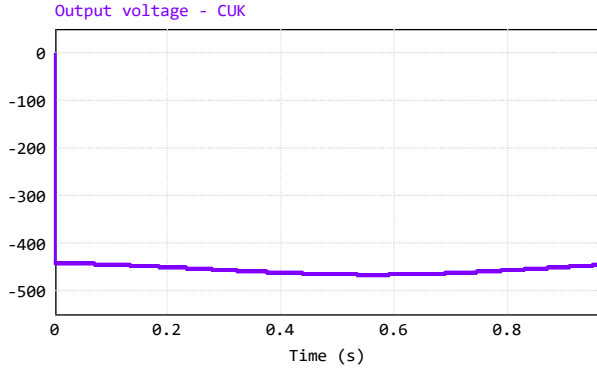


Fig. 15. Output voltage of the Cuk converter

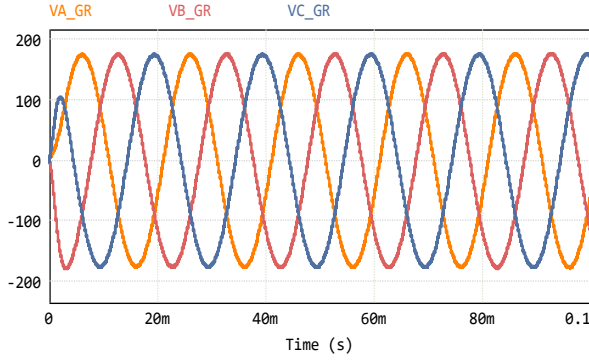


Fig. 16. Three-phase voltage obtained from the inverter

TABLE X
THREE-PHASE INVERTER VOLTAGE RESULTS

Magnitude	Expected (V_{RMS})	Simulated (V_{RMS})
Phase A–Neutral	127	128.161
Phase B–Neutral	127	128.355
Phase C–Neutral	127	128.189

TABLE XI
EXPECTED LOAD CURRENT VALUES

Period	RMS current (A)	Peak current (A)
00:00–06:00	0.333	0.4714045
06:00–11:00	5	7.0710678
11:00–16:00	20	28.284271
16:00–22:00	10	14.142136
22:00–24:00	0.333	0.4714045

E. Load Results

Only one inverter phase was used for the detailed load simulation in order to reduce simulation time. The selected single-phase voltage is shown in Fig. 17, and the current consumed by the load throughout the simulated day is shown in Fig. 18. Table XI lists the expected RMS and peak current values for each daily interval.

The simulated current waveform followed the scheduled load changes. The full-day current plot in Fig. 18 is kept as the representative load-current result; the individual period plots

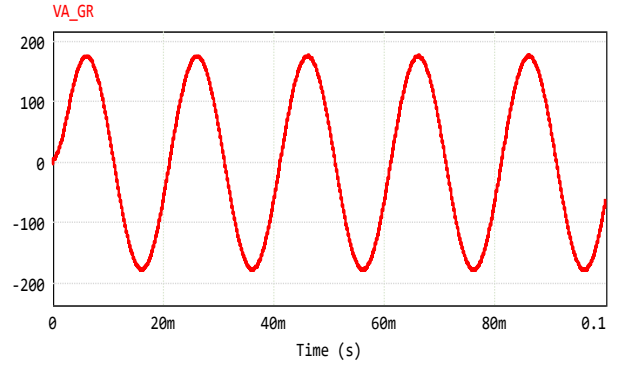


Fig. 17. Single-phase voltage selected to feed the load

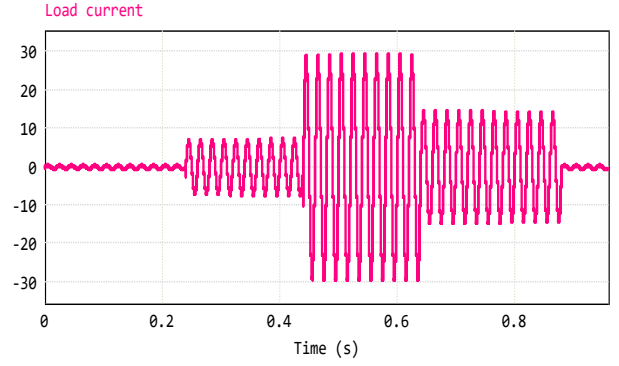


Fig. 18. Current consumed by the load during the simulated day

are omitted to avoid unnecessary repetition. The largest deviations were approximately 3%, which is acceptable considering the DC-bus ripple and the slight inverter-voltage increase.

VII. CONCLUSIONS

A stand-alone residential photovoltaic generation system with battery storage was designed and simulated. The PV farm, composed of 20 panels arranged as five series-connected modules in four parallel strings, achieved approximately 10 kW under peak irradiance. The Boost converter successfully adjusted the PV operating point and maintained the DC bus near 450 V, although manual duty-cycle changes introduced voltage perturbations.

The bidirectional Cuk converter fulfilled its role as the battery interface and generated the required bus voltage with inverted polarity. Small output fluctuations affected the downstream inverter but did not prevent satisfactory operation. The three-phase inverter produced phase voltages close to 127 V_{RMS}, and the load-current results matched the theoretical requirements with minor deviations.

Overall, the simulation demonstrates that the proposed isolated PV system can satisfy variable residential demand throughout the day while maintaining acceptable power quality. Future work should include closed-loop MPPT control, improved DC-bus regulation, and more advanced control strategies such as predictive, adaptive, or fuzzy-logic control for the bidirectional battery converter.

REFERENCES

- [1] C. Torres Vila, "Qué son los paneles solares, cómo funcionan y cuál es su futuro," *Scielo*, pp. 23–45, 2022.
- [2] Damia, "Conexión paneles solares en serie y en paralelo," Damia Solar Electrosol Energía S.L., pp. 1–4, 2016. [Online]. Available: https://www.damiasolar.com/actualidad/blog/articulos-sobre-la-energia-solar-y-sus-componentes/conexion-paneles-solares-en-serie-en-paralelo_1
- [3] Autosolar.pe, "¿Cómo es un regulador de carga MPPT?," pp. 2–5, 2016.
- [4] ENERGEMA, "Departamento de renovables: diodos bypass y de bloqueo en paneles fotovoltaicos," pp. 1–7, 2021.
- [5] L. A. D'Alessio, L. G. González, and R. O. Cáceres, "Diseño y construcción de un convertidor boost de pequeña potencia," *Avances en Ciencias e Ingeniería*, vol. 6, no. 2, pp. 2–7, 2014.
- [6] A. V. Ramos, "Convertidores CD-CD," Jun. 2014.
- [7] C. CD-CD, "Convertidores CD-CD 2.1," pp. 13–22.
- [8] A. A. Setiamey and E. Deliani, "No title," vol. 2, pp. 5–10, 2019.
- [9] I. I. Rahman, D. C. Riawan, and M. Ashari, "Design and implementation of DC-DC bidirectional Cuk converter with average current mode control for lead acid battery testing," in *Proc. 2019 Int. Seminar on Intelligent Technology and Its Application (ISITIA)*, 2019, pp. 183–188, doi: 10.1109/ISITIA.2019.8937200.
- [10] C. A. Ramos-Paja, A. J. Saavedra-Montes, and J. D. Bastidas-Rodríguez, "Cargador de baterías fotovoltaico con control por modos deslizantes y limitación de la derivada de corriente de carga," *Tecnológicas*, vol. 21, no. 42, pp. 129–145, 2018, doi: 10.22430/22565337.784.
- [11] M. O. No, F. A. C. Ial, and M. O. Dule, "Yangtze 210mm," pp. 1–2, 2018.